



An optimization framework for urban air mobility (UAM) planning and operations

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ABSTRACT

This paper presents an optimized decision-support framework for the planning and operation of Urban Air Mobility (UAM) systems. Alleviating traffic congestion in metropolitan areas has been a persistent challenge for decades, leading to increased interest in aerial mobility solutions. Recent advancements in distributed electric propulsion, battery technology, and autonomous navigation have made electric vertical take-off and landing (eVTOL) aircraft a feasible option for intercity transport. For efficient UAM systems, we optimize the high-level planning of UAMs, i.e., determine the numbers of eVTOLs, vertiport spaces, and chargers, together with lower-level operations to control each eVTOL's operational state between in-service, charging, idling, and relocating. Accounting for spatio-temporal demand and cost heterogeneity, we formulate the UAM optimization framework as a mixed-integer programming problem. In our numerical study, we analyze a scenario involving three hypothetical vertiports in the Seoul Metropolitan Area, South Korea. The results reveal relationships between the optimal solution and several exogenous factors critical to eVTOL operations, including the targeted level of service for users and eVTOL charging speed. Additionally, we conduct Monte Carlo simulations to demonstrate the robustness of our solution against stochastic demand and variations in electric consumption.

1. Introduction

Aerial transportation is increasingly recognized as a potential solution to alleviate ground traffic congestion in major metropolitan areas. Globally, millions of hours are lost daily due to traffic delays (Holden and Goel, 2016). This congestion significantly impacts productivity, environmental sustainability, and economic growth (Arnott and Small, 1994). Governments have endeavored to reduce traffic congestion through various policies. However, due to oversaturated ground infrastructure in urban areas, congestion remains a critical problem. This problem is leading governments and transportation service providers to seek a new, alternative mode: aerial-based mobility (Winter et al., 2020; Song and Yeo, 2021; Pons-Prats et al., 2022; Choi and Park, 2022).

In the past decade, three key technologies have matured, facilitating the development of Urban Air Mobility (UAM) services (Patterson et al., 2021; Tang et al., 2021): (i) distributed electric propulsion; (ii) batteries; and (iii) autonomous driving. Helicopters, introduced in the 1960s, have been used for a range of purposes, including weather observation, emergency transport, and firefighting. However, using helicopters for

urban mobility presents several challenges, such as excessive noise, high operational costs, and safety concerns, primarily due to the reliance on large single-engine combustion systems. With recent advancements, distributed electric propulsion systems equipped with smaller rotors than those used in helicopters now offer quieter, more efficient, and safer flights. Electric propulsion significantly reduces noise levels, while distributed motors provide redundancy, enhancing safety in the event of mechanical failures. Secondly, rapid advancements in battery technology, along with the increased adoption of electric vehicles, have led to lighter and more affordable electric batteries with higher energy density. This development allows UAM aircraft to operate more efficiently. Lastly, advancements in autonomous navigation and wireless communication technologies have improved the economic feasibility of UAM operations (Thipphavong et al., 2018; Cohen et al., 2021). With the integration of edge computing, aircraft can achieve greater control, agility, and cybersecurity, while centralized control centers can manage multiple aircraft simultaneously, ensuring system-wide efficiency.

Research on UAM operations has primarily focused on predicting UAM demand, identifying factors influencing UAM adoption, and

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determining optimal vertiport locations (Straubinger et al., 2020; Garrow et al., 2021; Goyal et al., 2021; Rajendran et al., 2021; Rath and Chow, 2022). Bulusu et al. (2021) proposed a traffic analysis method and estimated that up to 45% of demand could benefit from UAM, particularly during periods of high ground traffic congestion, such as morning and evening rush hours. Rimjha et al. (2021) analyzed UAM demand based on the number of vertiports and associated waiting times, employing a mode-choice conditional logit model. Stated preference surveys in several studies (Fu et al., 2019; Haddad et al., 2020; Peeta et al., 2008; Maheshwari et al., 2020; Ahmed et al., 2021) revealed that connectivity of UAM with other transportation modes, safety, and amount of travel time reduction were the essential factors in deciding whether to use UAM. Vascik and Hansman (2017) evaluated suitable vertiport locations, taking into account operational constraints such as public noise tolerance and the availability of suitable take-off and landing areas. Willey and Salmon (2021) solved a vertiport selection problem based on a p-median location problem. However, to the best of our knowledge, optimization of entire UAM systems, including planning and operational decision variables such as fleet size, vertiport spaces, chargers, and relocation strategies, has not yet been thoroughly explored.

In the taxonomy of mobility systems, UAM is classified as a one-way pod-based system in that trips begin and end at different designated stations (i.e., pods) (Maheshwari et al., 2020). One-way pod-based systems face operational challenges due to imbalanced vehicle distribution across pods, which arises from spatio-temporal demand variations (Nair and Miller-Hooks, 2011). This imbalance necessitates a vehicle relocation strategy in which, to satisfy demand, vehicles are dispatched to stations experiencing insufficient supply. Relocation strategies are typically categorized into user-based and operator-based models (Huang et al., 2020). For user-based models, Barth et al. (2004) proposed trip-joining and trip-splitting relocation strategies, which encourage users having the same origin and destination to join or split their trips according to the number of vehicles stored at the pick-up station. Furthermore, some studies suggested pricing incentives for users who choose their drop-off stations as those with vehicle shortages (Febbraro et al., 2012, 2019; Pfrommer et al., 2014). On the other hand, the operator-based model employs relocators to balance vehicle distribution. Some studies proposed a threshold-based relocation method that keeps the stock of vehicles within upper- and lower-bounds (Kek et al., 2009; Weikl and Bogenberger, 2015). When the number of vehicles exceeds the upper threshold, surplus vehicles are relocated to stations with fewer vehicles than the lower threshold. Gambella et al. (2018) suggested a dynamic vehicle relocation model based on a rolling horizon approach in which relocators employ real-time demand data to shift vehicles every short period ahead. One limitation of the operator-based model is that its computational complexity is significantly increased due to optimization of a single agent's detailed relocation strategy for multiple relocation staff members.

Several studies have optimized planning and operations for one-way pod-based systems, incorporating relocation strategies. Most methods have formulated Mixed Integer Programming (MIP) problems (Gambella et al., 2018; Jorge et al., 2014; Cepolina and Farina, 2012). Boyaci et al. (2015) introduced a multi-objective MIP model to simultaneously maximize net revenues for operators and users. They aggregated both planning decisions, such as station number and location, and fleet size, and operational decisions including vehicle relocation, assignment, and service trips, into a single formulation and solved it through a branch-and-bound approach. Shehadeh et al. (2021) proposed a two-stage stochastic MIP model to determine optimal fleet size and initial allocation for on-demand ground mobility services. The model aimed to minimize vehicle purchase costs and the expected total passenger waiting time under stochastic demand. Gambella et al. (2018) included consumption and recharge processes of electric vehicles in an MIP model. They compared an exact solution with one from a heuristic method that reduces the relocation density in terms of computing time

and performance. However, a major disadvantage of MIP-based models is their high computing cost for obtaining an exact solution, leading many studies to rely on heuristic methods, which may result in suboptimal solutions. More importantly, no previous study has optimized a one-way pod-based mobility system with relocations specifically tailored for UAM, while considering the distinct characteristics that differentiate UAM from traditional ground-based mobility systems. These differences include specialized stations, or vertiports, as well as the operational benefits and drawbacks of eVTOLs, which travel faster and avoid ground congestion but incur higher costs. To address this research gap, we propose an optimization model for planning and operations of UAM systems under spatio-temporal demand heterogeneity. We formulate the problem as an MIP, then decompose it into several smaller MIP problems, each corresponding to distinct time windows. This enables all decision variables to be independent of time sequence and, thus, significantly reduces computational time. We establish the constraints for the minimum required eVTOL fleet size, vertiport size, and charging infrastructure for a targeted level-of-service against demand stochasticity based on the workload transition-based model for on-demand mobility systems, suggested by Daganzo and Ouyang (2019). We expect that our solution can be a baseline for future UAM operation studies.

The remainder of this paper is structured as follows. First, we introduce the mechanism of UAM operation. Next, we present an MIP-based model and solution for operation strategy. Numerical study and results are described in the following section. We also conduct Monte Carlo simulations to observe a robustness of our proposed solution on stochastic demand and electric consumption. Finally, we present conclusions and limitations of this study.

2. UAM definitions

We introduce a mechanism for UAM systems and the basic assumptions used in this paper. The assumptions about the UAM operational environments while developing the proposed optimization model are as follows: The eVTOL aircraft in the UAM system are assumed to be capable of fully autonomous flight. Operations are primarily conducted under visual meteorological conditions, with the impacts of instrument meteorological conditions considered negligible for this study. The UAM traffic density is assumed to be moderate, based on predicted demand patterns, ensuring a balanced distribution of flights across the day to avoid airspace saturation. Urban airspace constraints, including no-fly zones around sensitive areas such as residential zones, hospitals, and government buildings, are taken into account, with airspace management adhering to existing regulations for safety and compliance. Public acceptance levels of UAM operations are assumed to be favorable, considering noise tolerance and fare acceptance based on survey data. These assumptions provide a realistic and comprehensive framework for the optimization model presented in this paper.

Let V be the set of N vertiports. The average UAM travel demand from vertiport $i \in V$ to vertiport $j \in V/\{i\}$ in each time period t is λ_{ij}^t , and the set of λ_{ij}^t for all time windows $t = 1, \dots, T$ in one day is $\lambda_{ij} = \{\lambda_{ij}^{t=0}, \lambda_{ij}^{t=1}, \dots, \lambda_{ij}^{t=T}\}$, $\forall i, j$. The length of each time period is Δ [hr]. We assume that vertiport i covers all UAM demands within its catchment area R_i . Users can use any kind of ground access mobility, such as walking and public transit, between their origins or destinations and the nearest vertiports. Fig. 1 conceptually shows the operation scenario of a UAM system with three vertiports.

In determining vertiport locations, we need to consider noise issues in residential areas, security issues, and connectivity with public transit. Even if electric propulsion technology reduces noise, constructing vertiports in residential areas might be avoided. One of the keys to increasing UAM users is that vertiports have to be positioned near transportation hubs (Straubinger et al., 2020; Rimjha et al., 2021). UAM

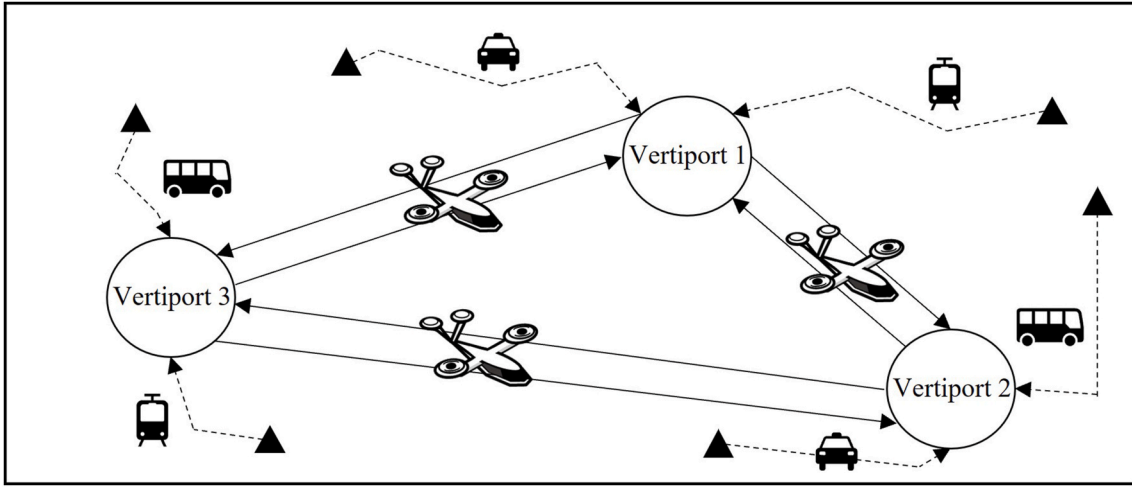


Fig. 1. A UAM scenario with three vertiports.

is a part of an intermodal transportation system, so user accessibility to ground transportation modes for first- and last-mile trips is essential. Such constraints mean that vertiport candidate locations are limited. Therefore, we assume that vertiports are already located at high-demand sites, such as transportation hubs and airports.

Understanding the mechanism of UAM operations is important to formulate the problem in a quantitative way. Fig. 2 shows the UAM system in graph form. The nodes represent the aircraft's current states, while the links indicate their state transitions. eVTOLs can be in one of three statuses: at a vertiport, serving, or relocating.

- At vertiport i : eVTOLs are idle, waiting, and charging at vertiport i
- Serving from vertiport i : eVTOLs are delivering passengers from origin vertiport i to a certain destination vertiport
- Relocating from vertiport i : eVTOLs are relocating to balance vehicle distribution from vertiport i to a certain destination vertiport

An eVTOL departs for another vertiport once it reaches full passenger capacity or when the maximum user waiting time exceeds a predefined threshold. After arriving at the destination vertiport, the eVTOL needs to stay there until its battery is fully recharged. The heavier the eVTOL, due

to the weight of motors, batteries, and passengers, the less efficient its electricity consumption, resulting in a reduced maximum travel range. For instance, eVTOLs that can accommodate four or more passengers have a maximum flight distance of fewer than 100 km (e.g., S-A1, Maker, and CityAirbus NextGen). Thus, reducing battery size to the minimum required to reliably cover the maximum vertiport-to-vertiport electricity consumption for all vertiport pairs and fully charging the battery between consecutive trips will be desirable, as shown in a study on optimal planning of electrified public transit systems by Lee et al. (2021). Once fully charged, the eVTOL is ready to be assigned to another trip or relocated.

3. Optimization model

3.1. Objective function

We classify UAM operating costs into two components: fleet operating costs and vertiport operating costs. Similar to the airline industry, where airlines lease dedicated portions of an airport and pay usage fees for the facilities provided by the airport operator—who is responsible for constructing and maintaining the airport infrastructure—we adopt a

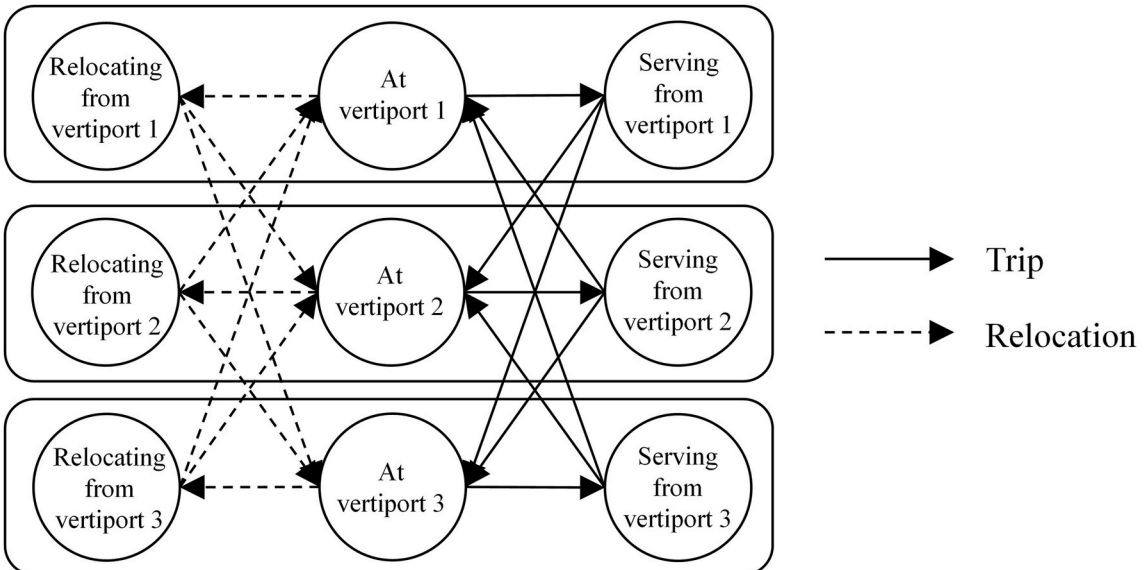


Fig. 2. Graph representation of UAM system for three zones.

similar framework for UAM operations. The objective of the proposed model is to minimize total operating costs from the perspective of the UAM fleet operator while guaranteeing a predefined level-of-service for users. The objective function is represented as:

$$\min_{m, x_i, y_i} J = c_1 \cdot m + c_2 \cdot E_c + \sum_i c_{3,i} \cdot x_i + \sum_i c_4 \cdot y_i. \quad (1)$$

where

- J : hourly operation cost [\$/hr], $\in \mathbb{R}^{\geq 0}$;
- c_1 : hourly prorated lifecycle eVTOL cost for purchasing and maintenance [\$/hr/vehicle], $\in \mathbb{R}^{> 0}$;
- m : number of the eVTOLs in system [vehicle], $\in \mathbb{N}_0$;
- c_2 : unit electricity price [\$/kWh], $\in \mathbb{R}^{> 0}$;
- E_c : hourly electric consumption [kWh/hr], $\in \mathbb{R}^{> 0}$;
- $c_{3,i}$: hourly average usage cost for single parking space at vertiport i [\$/space/hr], $\in \mathbb{R}^{> 0}$;
- x_i : number of parking spaces dedicated to the UAM operator at vertiport i [space], $\in \mathbb{N}_0$;
- c_4 : hourly average usage cost for a single charger, excluding the cost for charged electricity [\$/charger/hr], $\in \mathbb{R}^{> 0}$; and
- y_i : number of required chargers dedicated to the UAM operator at vertiport i [charger], $\in \mathbb{N}_0$.

J denotes prorated daily operation cost, consisting of fleet purchase and maintenance cost, eVTOL operation cost, space cost at vertiport, and charger installation and operation cost. Decision variables are set to the number of VTOLs in the entire system, m , and the number of parking spaces and chargers at each vertiport, x_i and y_i for all i , respectively. The unit parking space cost, $c_{3,i}$, varies among vertiports because the price of real estate depends highly on locations of vertiports. Other cost factors for mandatory system components, such as landing pads and infrastructure for passengers, are not included, because they are not decision factors but requirements.

3.2. Constraints

Little's (1961) proved that the total number of customers in a system is equal to the arrival rate multiplied by the average time a customer spends in that system. In the UAM system, the required fleet size to meet all demand within time window t can be derived by multiplying flight duration time by the flight departure rates for service and relocation. The fleet size, m , must be larger than or equal to the sum of eVTOLs in service, charging, relocating, and staying at vertiports in all time windows $t = 1, \dots, T$. Based on Little's Law, the minimum required fleet size is formulated as follows:

$$m \geq \sum_{ij} n_{ij}^t \cdot \left(\tau_{ij} + \frac{ec_{s,ij}^t}{\gamma} \right) + \sum_{ij} r_{ij}^t \cdot \left(\tau_{ij} + \frac{ec_{r,ij}^t}{\gamma} \right) + \sum_i w_i^t, \forall t = 1, \dots, T. \quad (2)$$

where

- n_{ij}^t : eVTOL flight frequency for service trips from vertiport i to j in time window t [vehicle/hr], $\in \mathbb{R}^{\geq 0}$;
- τ_{ij} : flight time from vertiport i to j [hr], $\in \mathbb{R}^{> 0}$;
- r_{ij}^t : eVTOLs flight frequency for relocation trips from vertiport i to j in time window t [vehicle/hr], $\in \mathbb{R}^{\geq 0}$;
- $ec_{s,ij}^t$: energy consumption for in-service flight from vertiport i to j in time window t [kWh], $\in \mathbb{R}^{> 0}$;
- $ec_{r,ij}^t$: energy consumption for relocation flight from vertiport i to j [kWh], $\in \mathbb{R}^{> 0}$;
- w_i^t : number of idle eVTOLs staying at vertiport i in time window t [vehicles], $\in \mathbb{N}_0$;
- γ : charging speed [kW], $\in \mathbb{R}^{> 0}$;
- $\sum_{ij} n_{ij}^t \cdot \tau_{ij}$: number of eVTOLs in service trips in time window t [vehicles], $\in \mathbb{R}^{\geq 0}$;

- $\sum_{ij} r_{ij}^t \cdot \tau_{ij}$: number of relocating eVTOLs in time window t [vehicles], $\in \mathbb{R}^{\geq 0}$;
- $\sum_{ij} \frac{ec_{s,ij}^t \cdot n_{ij}^t + ec_{r,ij}^t \cdot r_{ij}^t}{\gamma}$: number of eVTOLs charging at all vertiports in time window t [vehicles], $\in \mathbb{R}^{\geq 0}$; and
- $\sum_i w_i^t$: number of idle eVTOLs at all vertiports in time window t [vehicles], $\in \mathbb{N}_0$.

The electric consumption of eVTOL is weight-dependent, resulting in different electric consumptions for in-service flights, $ec_{s,ij}^t$, and relocation flight, $ec_{r,ij}^t$, based on the average number of onboard passengers. Note that a relocating eVTOL does not carry passengers, leading to a lower and time-consistent $ec_{r,ij}^t$ compared to $ec_{s,ij}^t$. Charging time can be calculated by dividing the energy consumption to fly from vertiport i to j by the charging speed of a charger, γ . As eVTOLs must be fully charged after trip and relocation flights, the total duration between take-off at vertiport i for a flight from i to j and the state of being ready at vertiport j for the next flight is the summation of flight time τ_{ij} and the charging time after the flight, $\frac{ec_{s,ij}^t}{\gamma}$ or $\frac{ec_{r,ij}^t}{\gamma}$.

The average departure rate of service trips from vertiport i to j in time window t , n_{ij}^t , is $\frac{\lambda_{ij}^t R_i}{\kappa}$, where κ indicates the capacity of one eVTOL in units of [passengers/vehicle]. When the average period between consecutive departures from vertiport i to j , $WT_{ij}^t \equiv \frac{\kappa}{\lambda_{ij}^t R_i}$, is over the threshold $WT_{\max}$, predetermined to avoid any overly long waiting time for early-arrived passengers, i.e., $WT_{ij}^t \leq WT_{\max}$, n_{ij}^t is the minimum allowed frequency $\frac{1}{WT_{\max}}$. In summary, n_{ij}^t is:

$$n_{ij}^t = \min \left\{ \frac{\lambda_{ij}^t R_i}{\kappa}, \frac{1}{WT_{\max}} \right\}, \forall i, j, t. \quad (3)$$

The airspeed of an eVTOL can be divided into seven flight phases, excluding two loitering phases (Beyne and Castro, 2022). The horizontal airspeed graph with the flight phases is represented in Fig. 3. In phases 1 and 7, eVTOLs vertically take off and land, respectively. In phase 4, eVTOL cruises at maximum flight speed. In the other phases, the aircraft accelerates or decelerates while ascending or descending until it reaches cruise level and landing level, respectively.

Total travel time from vertiport i to j , τ_{ij} , is divided into cruising, take-off, and landing times, as in Equation (4). Take-off and landing times are affected by height of vertiport and type of eVTOL. Otherwise, total travel time can also be approximately calculated as $\tau_{ij} = \frac{d_{ij}}{v_{\text{avg}}}$, where d_{ij} is the air travel distance between vertiports i and j , and v_{avg} is the average airspeed, which is lower than the cruise speed.

$$\tau_{ij} = \tau_{ij}^c + (\tau^o + \tau^l), \forall i, j, \quad (4)$$

where

- τ_{ij} : total travel time from vertiport i to j [hr];
- τ_{ij}^c : cruising time from vertiport i to j (phase 4) [hr];
- τ^o : take-off time (phases 1, 2 and 3) [hr]; and
- τ^l : landing time (phases 5, 6 and 7) [hr].

The electric consumption efficiency varies between the cruising, take-off, and landing phases. For an eVTOL with full passenger occupancy, the sum of electricity consumption for take-off and landing is ec_1 [kWh]. Electric consumption in the cruising phase is proportional to the amount of time spent cruising, τ_{ij}^c , if the cruise speed, v_c , is fixed. The power required to cruise is ec_2 [kW]. There is a trade-off between hovering and cruising efficiency. For example, vectored thrust eVTOLs have wings for efficient cruising but have poor hovering efficiency; wingless eVTOLs (e.g., multicopter) have large disk actuator surfaces for efficient hovering but show low cruise efficiency.

Weight-dependent energy consumption is primarily determined by the number of passengers onboard. To calculate the electricity consumption for in-service and relocation flights, we first define the relative

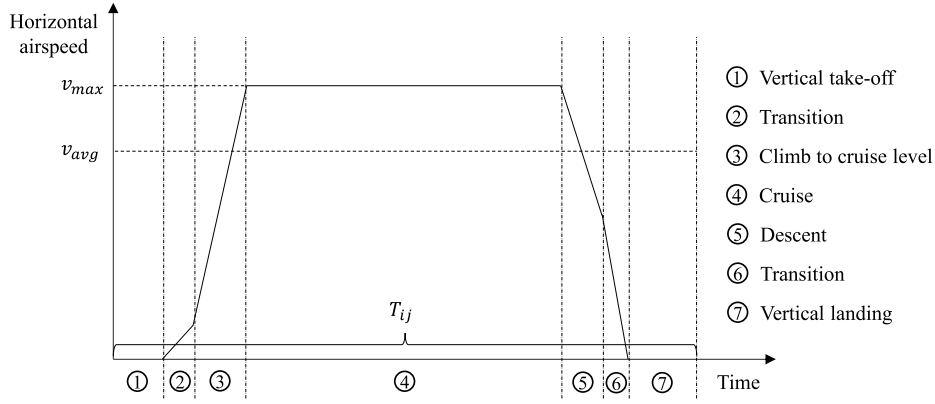


Fig. 3. Horizontal airspeed graph of eVTOL (Beyne and Castro, 2022).

eVTOL weight ratios to the full eVTOL as Equations (5) and (6), respectively.

$$pw_{s,ij}^t = \frac{em + pm \cdot \frac{\lambda_{ij}^t R_i}{n_{ij}^t}}{em + pm \cdot \kappa}, \forall i, j, t \text{ and} \quad (5)$$

$$pw_{r,ij} = \frac{em}{em + pm \cdot \kappa}, \forall i, j, t. \quad (6)$$

where

$pw_{s,ij}^t$: average weight ratio for in-service from vertiport i to j in time window t ;

$pw_{r,ij}$: weight ratio for relocation from vertiport i to j ;

em : eVTOL weight without any passenger [kg]; and

pm : average weight per a passenger [kg/passenger].

To ensure reliable operation, we account for the inherent stochasticity in electricity consumption, caused by multiple factors including adverse weather; we parameterize this stochasticity with the mean-to-standard-deviation ratio α [kWh/kWh]. Finally, based on the weight ratios, the total electricity consumptions for in-service and relocation flights with a confidence interval of q_1 (e.g., $\Phi^{-1}(q_1) = 1.96$ for $q_1 = 97.5\%$ of CI) are represented as:

$$ec_{s,ij}^t = (ec_1 + ec_2 \cdot \tau_{ij}^c) \cdot pw_{s,ij}^t + \Phi^{-1}(q_1) \alpha \sqrt{(ec_1 + ec_2 \cdot \tau_{ij}^c) \cdot pw_{s,ij}^t}, \forall i, j, t \text{ and} \quad (7)$$

$$ec_{r,ij} = (ec_1 + ec_2 \cdot \tau_{ij}^c) \cdot pw_{r,ij} + \Phi^{-1}(q_1) \alpha \sqrt{(ec_1 + ec_2 \cdot \tau_{ij}^c) \cdot pw_{r,ij}}, \forall i, j. \quad (8)$$

A sufficient number of idle eVTOLs must be maintained at each vertiport to allow them to be assigned to passengers without significantly excessive waiting time. Similar to the method of Daganzo and Ouyang (2019), we assume that each vertiport has at least one idle vehicle; we add a buffer for resilience against stochastic demand, which is defined with another confidence level q_2 and the mean-to-variance ratio β [vehicle/hr]:

$$w_i^t \geq 1 + \Phi^{-1}(q_2) \sum_j \sqrt{2n_{ij}^t \beta}, \forall i, t. \quad (9)$$

If $\beta = 1$, departures of eVTOLs follow a Poisson process. If $n_{ij}^t = \frac{1}{WT_{\max}}$, and the period between departures is fixed at $WT_{\max}$, β is zero.

The required number of parking spaces at vertiport i is determined by the maximum number of eVTOLs charging or idle for all time windows, as follows:

$$x_i \geq \max_t \left\{ w_i^t + \sum_j \frac{ec_{s,ij}^t \cdot n_{ij}^t + ec_{r,ij} \cdot r_{ji}^t}{\gamma} \right\}, \forall i. \quad (10)$$

The required number of chargers at vertiport i is the maximum number of eVTOLs charging for all time windows, as follows:

$$y_i \geq \max_t \left\{ \sum_j \frac{ec_{s,ij}^t \cdot n_{ij}^t + ec_{r,ij} \cdot r_{ji}^t}{\gamma} \right\}, \forall i. \quad (11)$$

Last, relocation of eVTOLs is used to balance the stock of each vertiport. Thus, we need the following conservation constraint:

$$\sum_j n_{ji}^t - \sum_j n_{ij}^t = \sum_j r_{ij}^t - \sum_j r_{ji}^t, \forall i, t. \quad (12)$$

3.3. Solution methodology

To reduce the problem complexity, we decompose the original problem into T steady-state problems defined for each discrete time window. We assume that the duration of each time window, Δ , is significantly longer than the maximum travel time and charging time for a flight (i.e., $\Delta \gg \tau_{ij}^c$ and $\frac{ec_{s,ij}^t}{\gamma}$ and $\frac{ec_{r,ij}}{\gamma} \forall i, j, t$), allowing us to ignore the transient period between the steady states of two consecutive time windows.

The objective function of the steady-state problem in time window t is as follows:

$$\begin{aligned} \min_{r_{ij}^t, w_i^t, x_i^t, y_i^t} \quad & c_1 \cdot \Delta \cdot \left(\sum_{ij} n_{ij}^t \cdot \left(\tau_{ij} + \frac{ec_{s,ij}^t}{\gamma} \right) + \sum_{ij} r_{ij}^t \cdot \left(\tau_{ij} + \frac{ec_{r,ij}}{\gamma} \right) + \sum_i w_i^t \right) \\ & + c_2 \cdot \Delta \cdot \left(\sum_{ij} n_{ij}^t \cdot ec_{s,ij}^t + \sum_{ij} r_{ij}^t \cdot ec_{r,ij} \right) \\ & + \Delta \cdot \sum_i c_{3,i} \cdot x_i^t + \Delta \cdot \sum_i c_{4,i} \cdot y_i^t. \end{aligned} \quad (13)$$

We use the minimum required fleet size, the right-hand side of Constraint (2), as the fleet size, so we do not need to set the fleet size as a decision variable; rather, the relocation rates r_{ij}^t and the number of idle vehicles w_i^t are selected as decision variables instead. The required numbers of parking spaces and chargers at vertiport i in time window t are denoted by x_i^t and y_i^t , respectively.

The following constraints are considered in this sub-level problem, subject to the following:

$$w_i^t \geq 1 + \Phi^{-1}(q_2) \sum_j \sqrt{2n_{ij}^t \beta}, \forall i, t; \quad (9)$$

$$\sum_j n_{ji}^t - \sum_j n_{ij}^t = \sum_j r_{ij}^t - \sum_j r_{ji}^t, \forall i, t; \quad (12)$$

$$x_i^t \geq w_i^t + \sum_j \frac{ec_{s,ij}^t \cdot n_{ji}^t + ec_{r,ji}^t \cdot r_{ji}^t}{\gamma}, \forall i; \quad (14)$$

$$y_i^t \geq \sum_j \frac{ec_{s,ij}^t \cdot n_{ij}^t + ec_{r,ij}^t \cdot r_{ij}^t}{\gamma}, \forall i; \quad (15)$$

$$r_{ij}^t \in \mathbb{R}^{\geq 0}, \forall i, j; \text{ and} \quad (16)$$

$$w_i^t, x_i^t, y_i^t \in \mathbb{N}_0, \forall i. \quad (17)$$

We adopt Constraints (9) and (12) from the previous section, but Constraints (10) and (11) are modified to Constraints (14) and (15) using x_i^t and y_i^t instead of x_i and y_i . Constraints (16) and (17) define the domains of solutions. The optimal values of r_{ij}^t and w_i^t represent the optimal operations of relocation and idling for time window t . The operations for servicing passenger trips and charging are provided exogenously. The above sub-level problem is a Mixed-Integer Linear Programming (MILP) model.

At the planning level, we use the optimal solution for each sub-level problem (r_{ij}^{t*} , w_i^{t*} , x_i^{t*} , and y_i^{t*} for all i, j , and t) to determine the optimal number of eVTOLs, parking spaces, and chargers at each vertiport. We select the highest values from the optimal solutions across all sub-level problems for T time windows:

$$m^* = \max_t \left\{ \sum_{ij} n_{ij}^t \cdot \left(\tau_{ij} + \frac{ec_{s,ij}^t}{\gamma} \right) + \sum_{ij} r_{ij}^{t*} \cdot \left(\tau_{ij} + \frac{ec_{r,ij}^t}{\gamma} \right) + \sum_i w_i^{t*} \right\}; \quad (18)$$

$$x_i^* = \max_t x_i^{t*}, \forall i; \text{ and} \quad (19)$$

$$y_i^* = \max_t y_i^{t*}, \forall i. \quad (20)$$

If eVTOLs are not in operation at night, the total number of parking spaces should equal the fleet size m . When the parking space cost is even higher than the relocation cost, it is economically beneficial to move excess eVTOLs to vertiport $k \in V$ associated with the lowest unit parking

costs after daily operation. In this case, the optimal number of parking spaces is:

$$x_k^* = \max_t x_k^{t*}, \forall k \in V \setminus \{k\}; \text{ and} \quad (21)$$

$$x_k^* = m^* - \sum_{i \in V \setminus \{k\}} \left(\max_t x_i^{t*} \right). \quad (22)$$

4. Numerical example

4.1. Study site

We select vertiport locations near Seoul train station, Incheon Airport, and Dongtan train station in South Korea, strategically positioned near other modes of transit. Each vertiport's location and distances from the other vertiports are presented in Fig. 4. For this study, we assume that airways are set to follow the shortest distances between two vertiports.

4.2. Cost and operational parameters

The cost and operational parameters used in this case study are outlined in Table 1. The purchase cost of eVTOLs can vary significantly depending on manufacturer and design. For example, Joby Aviation's Joby S4, with a four-seat capacity, is estimated to cost \$1.3 million to manufacture (Vertical Flight Society, 2021). According to L.E.K Consulting (2021), in the early stages of a UAM system, the cost of a five-seater eVTOL could exceed \$4 million. We assume our eVTOLs have a four-passenger capacity and a purchasing cost of approximately \$3 million. The building cost of vertibases, estimated by McKinsey & Company (2021) to range between \$500,000 and \$800,000, varies by location. We amortize the costs of purchasing eVTOLs and building vertiport over ten years with a 2% interest rate. We assume the vertiport space usage cost is proportional to real estate prices, with the highest cost at Seoul Station and the lowest at Incheon Airport.

Holden and Goel (2016) suggest long-distance service can be

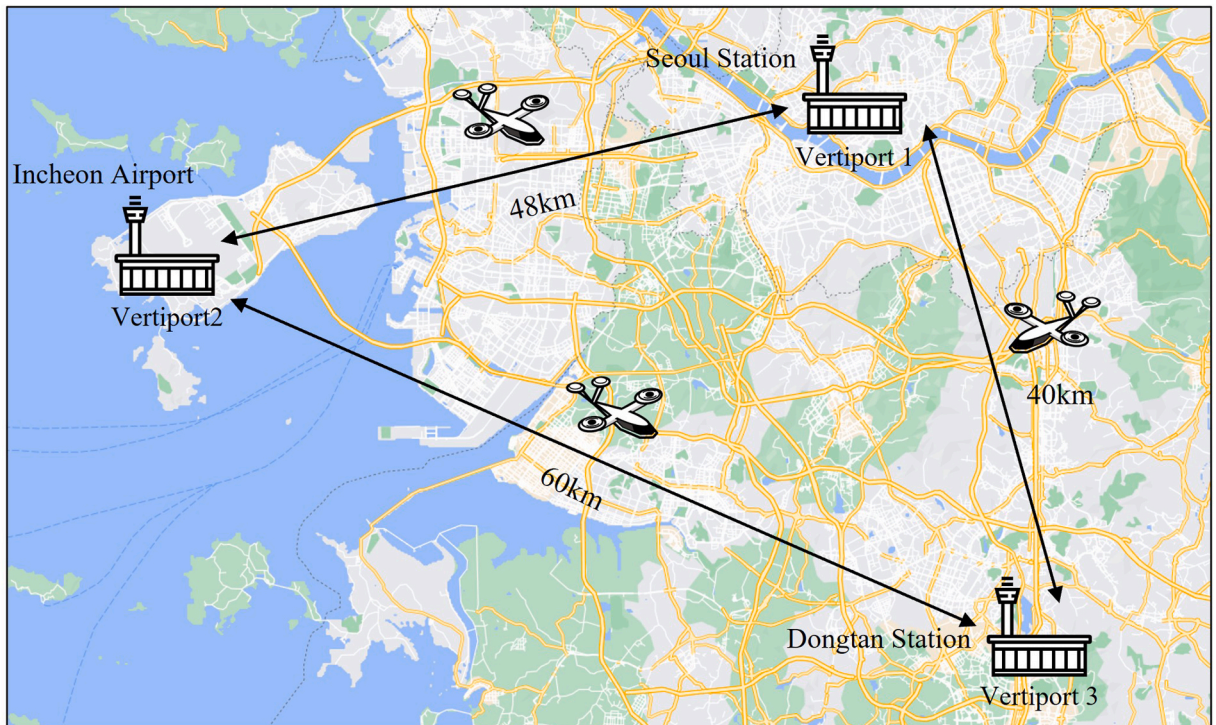


Fig. 4. Study site and example of three vertiport locations: Seoul station; Incheon airport; and Dongtan station.

Table 1
Cost and operational parameters used in numerical example.

Notation	Meaning	Value	Units	Reference
c_1	eVTOL cost	76.28	\$/aircraft/hr	L.E.K. Consulting (2021); Vertical Flight Society (2021)
c_2	Electric cost	0.12	\$/kWh	Holden and Goel (2016)
$c_{3,1}, c_{3,2}, c_{3,3}$	Vertiport space usage cost	29.51, 19.08, 20.99	\$/space/hr	McKinsey & Company, 2021
c_4	Charger usage cost	4.57	\$/charger/hr	Goyal et al. (2018)
v_c	Cruising airspeed	240	km/hr	Holden and Goel (2016)
τ^o	Take-off time	0.017	hr	Holden and Goel (2016)
τ^l	Landing time	0.017	hr	Holden and Goel (2016)
τ_{12}^c	Cruising time between Vertiports 1 and 2	0.2	hr	–
τ_{23}^c	Cruising time between Vertiports 2 and 3	0.25	hr	–
τ_{13}^c	Cruising time between Vertiports 1 and 3	0.17	hr	–
ec_1	Electric consumption for take-off and landing	16.7 ^a	kWh	Holden and Goel (2016)
ec_2	Electric consumption for cruising	71 ^a	kW	Holden and Goel (2016)
γ	Charging speed	150	kW	Goyal et al. (2018)
Δ	Time window	2	hr	–
T_o	Total operation time	12	hr	–
T	Total number of time windows	6	–	–
$\alpha, \forall i, j$	Mean-to-standard-deviation for electric consumption	0.1	kWh/kWh	–
$\beta, \forall i, t$	Mean-to-variance for stochastic demand	0.1	vehicle/hr	–
q_1	Confidence interval for electric consumption	0.95	–	–
q_2	Confidence interval for stochastic demand	0.95	–	–
$WT_{\max}$	Maximum waiting time	10	min	–
κ	eVTOL capacity	4	passengers	–
em	eVTOL weight without any passenger	1,574	kg	Holden and Goel (2016)
pm	Average weight per a passenger	85	kg	–

^a The values of represented electric consumption are calculated when the capacity of eVTOL is full (4 passengers).

economically feasible with a maximum one-minute take-off and landing (i.e., $\tau^o = \tau^l = 0.017$ hours). The total daily operating time is set to 12 hours, from 7 a.m. to 7 p.m., with each time window being 2 hours. Thus, $T_o = 12$ hours, and $\Delta = 2$ hours. The travel time between Incheon Airport and Dongtan Station is approximately 0.25 hours, with an additional 0.23 hours required for battery recharging after the trip.

4.3. UAM demand

The Korea Transportation Database provides hourly transit demand for each region; because accurate UAM demand survey analysis has not yet been conducted for the three locations, we multiply an arbitrarily chosen proportion by the demand for each fastest mode to construct an example of UAM demand. Table 2 shows the expected UAM demand.

4.4. Optimization results

The optimization model was implemented using MATLAB, employing the 'intlinprog' solver to specifically address the MILP sub-level problem. Solver parameters were configured with a relative gap tolerance of 10^{-4} and an absolute gap tolerance of 10^{-6} . All numerical studies were conducted on an Apple M3 Pro computer with a 36 GB RAM. The computation time was measured as less than one second.

Table 3 shows the optimal numbers of eVTOLs, m_i^* , parking spaces, x_i^* , and chargers, y_i^* at each vertiport for all time windows. In this case, the optimal planning is mainly determined by the solution for the time window of $t = 6$. Since vertiport 2 has the lowest parking space cost, the additional number of parking spaces needed to store all eVTOLs overnight is added to the optimal number of parking spaces, x_2^* .

The optimal flight frequency for relocation, r_{ij}^* , and the number of idle eVTOLs, w_i^* , are presented in Table 4. At $t = 3$, relocation flights do not occur since all pairs of service trip frequencies are constrained to the minimum frequency. Because we set the maximum waiting time, $WT_{\max}$, to 10 minutes, the minimum flight frequency has to be larger than six trips per hour.

Each cost component for planning and operating UAM systems is presented in Fig. 5. According to McKinsey & Company (2021), the annual operating cost for a vertiport with nine spaces ranges from \$3 million to \$5 million. This cost includes rent, land use, labor, and traffic management. Assuming the operation cost is linearly proportional to the number of vertiport spaces, we derived the daily vertiport operation cost. We found that vertiport operation costs account for 49% of the total operating costs, while fleet costs constitute over one-third of the total cost. In contrast to the existing airline industry, where fuel accounts for a substantial portion of the total cost (Singh and Sharma, 2015), the electric fuel cost in UAM systems is only 2%. This suggests that, despite frequent relocations after service hours, allocating additional parking spaces to the vertiport with the lowest cost is economically advantageous.

This study assumes autonomous flight, hence pilot labor costs are not included. According to the U.S. Bureau of Labor Statistics (2023), the average annual salary for commercial pilots is approximately \$113,080. Assuming that the required number of pilots equals the fleet size, the daily pilot labor cost would be \$24,792, increasing the total cost by 26%.

4.5. Sensitivity analysis

In this section, we conduct a comprehensive sensitivity analysis to explore how variations in key operational and cost parameters affect the daily UAM operation cost. We systematically adjust the costs of eVTOLs, vertiport spaces, and operational settings in our models to assess their effects on the overall cost-effectiveness and operational efficiency of UAM systems. By altering key cost parameters, this analysis evaluates the robustness of our operational strategies under different economic conditions, guiding stakeholders in effective UAM deployment.

Figs. 6 and 7 illustrate that daily operation costs and the minimum required fleet size decrease nearly exponentially with respect to charging speed. Charging speed reduces not only the required number of eVTOLs but also the number of chargers and parking spaces. The U.S. Department of Energy has invested in the Battery 500 project, which aims to increase the charging speed to 350kW (Ping and Shirley, 2022). If the charging speed is improved from 150kW to 350kW, operators can

Table 2

Expected UAM demand [passengers/hour].

Time	$\lambda_{12}^t R_1$	$\lambda_{13}^t R_1$	$\lambda_{21}^t R_2$	$\lambda_{23}^t R_2$	$\lambda_{31}^t R_3$	$\lambda_{32}^t R_3$	Total demand
$t = 1$ (7 ~ 9)	42	30	21	11	47	16	167
$t = 2$ (9 ~ 11)	18	37	9	13	34	15	126
$t = 3$ (11 ~ 13)	8	24	16	11	11	4	74
$t = 4$ (13 ~ 15)	8	32	16	16	22	7	101
$t = 5$ (15 ~ 17)	11	40	21	15	27	12	126
$t = 6$ (17 ~ 19)	26	47	53	15	29	6	176

Table 3

Minimum Required Parking Spaces, Chargers, and eVTOLs According to Discrete Time Windows.

Time	Parking spaces			Chargers			eVTOL aircraft
	x_1^*	x_2^*	x_3^*	y_1^*	y_2^*	y_3^*	m^*
$t = 1$ (7 ~ 9)	9	5	9	4	4	4	33
$t = 2$ (9 ~ 11)	8	4	8	4	3	4	27
$t = 3$ (11 ~ 13)	4	4	4	3	3	3	19
$t = 4$ (13 ~ 15)	7	4	4	3	3	3	23
$t = 5$ (15 ~ 17)	8	4	8	4	3	4	27
$t = 6$ (17 ~ 19)	10	10	8	5	5	4	38
Required ($x_1^*, y_1^*, m^*, \forall i$)	10	10 + 9 ^a	9	5	5	4	38

^a additionally required parking spaces for nighttime parking.

save \$8,870 per day, 18% of the total agency cost when the charging speed is 150kW.

Fig. 8 represents the variation in daily operation costs as a function of eVTOL cruising speed, taking into account the presence of no-fly zones and high-rise buildings that necessitate route deviations. Additionally, air congestion due to high eVTOL traffic may further slow average cruising speeds. Our analysis reveals that operational costs decrease exponentially with increases in cruising speed. Specifically, the cost reduction associated with increasing the cruising speed from 140km/h to 240km/h is \$14,381 per day, while a speed increase from 240km/h to 340km/h results in a reduction of \$6,095 per day. These findings highlight the critical importance of optimizing air routes to minimize deviations and maximize average cruising speed, suggesting that while advancements in eVTOL technology are beneficial, efficient route management may yield more significant cost savings.

The operation cost has a trade-off relationship with the maximum allowable waiting time, WT_{max} . Reducing the maximum waiting time improves level-of-service but also raises the operation cost because of additional trips. As shown in Fig. 9, the operation cost decreases exponentially up to a particular range of maximum allowable waiting time and remains constant after that. This is because the number of additional flights required to maintain the LOS converges to nearly zero. In other words, after that range, most users' waiting times are shorter than the

maximum allowable waiting time. In addition, we found that the range moves according to total demand. When total demand decreases to 90%, the maximum range is extended from approximately 13.5 minutes to 18 minutes.

Fig. 10 illustrates the influence of varying eVTOL and vertiport space costs on daily operational expenses, demonstrating a linear reduction in costs as inputs decrease from 150% to 50% of baseline values specified in Table 1. Notably, a 10% reduction in both eVTOL and vertiport costs leads to daily operational savings of \$4,265 and \$1,333 respectively, underscoring the critical impact of cost management on the economic viability of UAM systems. These findings suggest that strategic cost control, possibly through negotiations on eVTOL purchases and vertiport leases, alongside the adoption of cost-effective technologies and operational efficiencies, could significantly enhance the sustainability and adoption of UAM services.

We ran our model 100 times to evaluate the average computation time for different numbers of vertiports, as depicted in Fig. 11. For each simulation, we varied the hourly UAM demand between 5 and 50 and the distance between vertiports from 50 to 100km. Despite the model's NP-hard nature, the computation times were exceptionally fast, typically under 2 seconds for scenarios with up to 15 vertiports. This efficiency is due to our approach of decomposing the complex problem into several steady-state subproblems, significantly reducing the number of variables and constraints. For a system with an even larger number of vertiports, it is more efficient to treat the integer decision variables, such as the number of parking spaces and chargers, as continuous decision variables. This reduces the problem's complexity to polynomial time. Although the solution may include a safety buffer (i.e., a higher objective value with lower failure rates), the final integer decisions can be obtained by rounding up the continuous optimal results.

5. Agent-based modeling and simulations

In this section, we conduct agent-based Monte Carlo simulations to validate the robustness of our proposed solution against stochastic demands and variable electric consumption. We define eVTOLs, operated by UAM operators, and passengers as agents with distinct roles: eVTOLs handle route selections and operational adjustments; passengers influence system demand through their travel timing decisions. The planning variables, such as the number of spaces and chargers at each vertiport

Table 4

Optimal Relocation Frequency and Number of Idle eVTOLs According to Discrete Time Windows.

Time	Relocation frequency						Number of idle aircraft		
	r_{12}^*	r_{13}^*	r_{21}^*	r_{23}^*	r_{31}^*	r_{32}^*	w_1^*	w_2^*	w_3^*
$t = 1$ (7 ~ 9)	0	0	0.25	4.25	0	0	5	1	5
$t = 2$ (9 ~ 11)	0	0	0	0	0.75	0	4	1	4
$t = 3$ (11 ~ 13)	0	0	0	0	0	0	1	1	1
$t = 4$ (13 ~ 15)	0	0	0	0	2	0	4	1	1
$t = 5$ (15 ~ 17)	0	0	0	0	3.25	0	4	1	4
$t = 6$ (17 ~ 19)	2.25	0	0	0	0	4.5	5	5	4

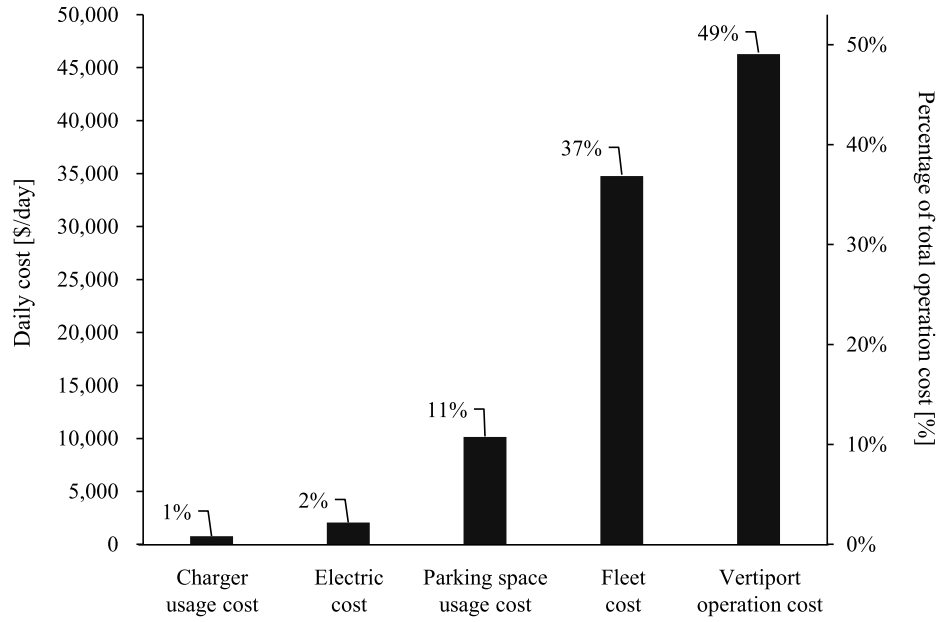


Fig. 5. Cost components for planning and operation of UAM systems.

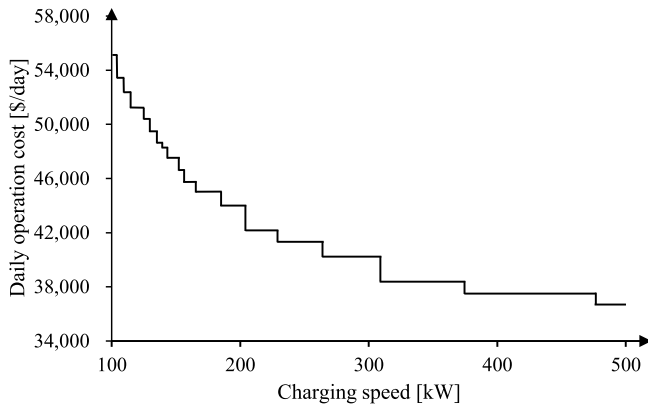


Fig. 6. Daily operation cost with respect to charging speed.

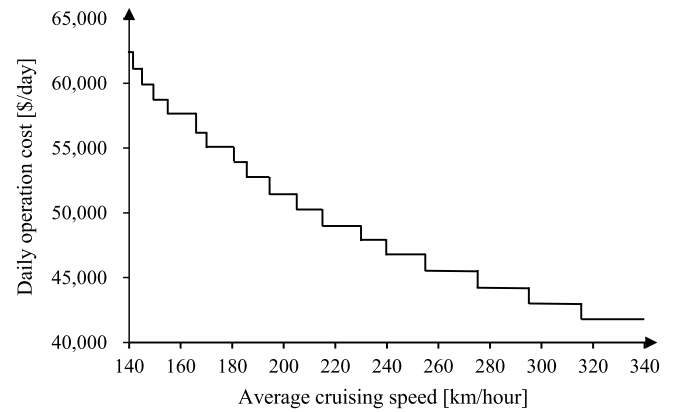


Fig. 8. Daily operation cost under different eVTOL cruising speed.

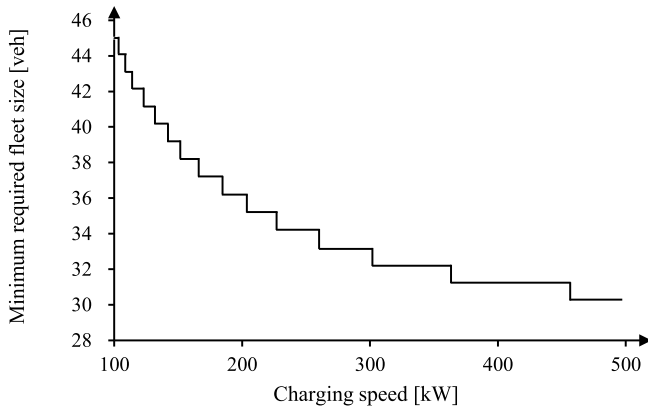


Fig. 7. Minimum required fleet size with respect to charging speed.

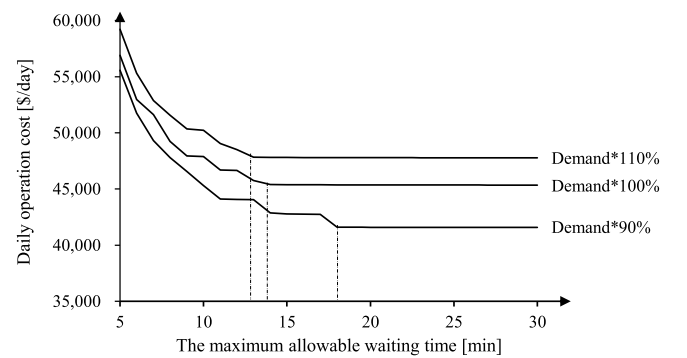


Fig. 9. Daily operation cost with respect to maximum allowable waiting time.

and the fleet size, are fixed based on the optimal values obtained from our previous analyses. This setup tests the robustness of our operational plans under realistic and unpredictable conditions, mirroring the randomness of passenger arrivals at vertiports.

We applied nearly the same operational rules: eVTOLs must be fully

charged, and trips begin either when the eVTOL is at full capacity or when the maximum allowable waiting time has been exceeded. However, in real-world operations, demand-responsive relocations are necessary based on real-time circumstances, while at the planning stage, the average relocation flows were taken into account instead. In the Monte Carlo simulations, we adopt the following optimal relocation strategy. We set the relocation time interval to ten minutes, and the

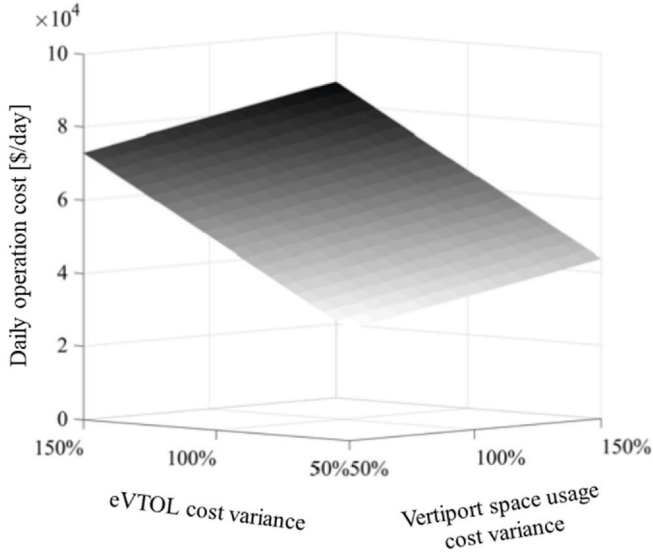


Fig. 10. Daily operation cost under different variances of eVTOL and vertiport space cost.

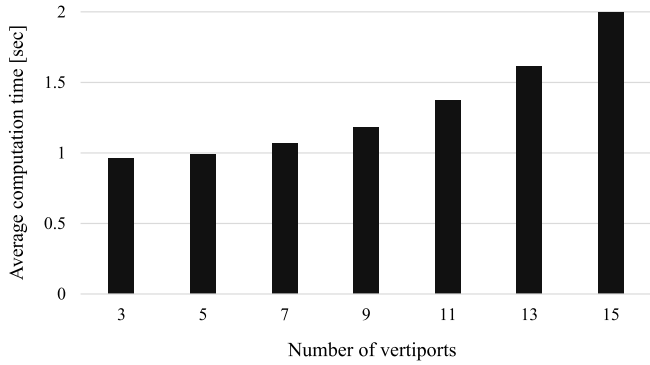


Fig. 11. Average computation time under different number of vertiports.

relocation time index is denoted by $t_r = 1, \dots, T_r$. For $t_r = 1$, it is unnecessary to relocate eVTOLs. For $t_r = 2, \dots, T_r$, the required relocation trips are calculated by Equations (23)–(25). Equation (23) is the objective function to minimize the electric costs for relocations. The actual number of relocating eVTOLs from vertiport i to vertiport j at t_r is denoted $RL_{ij}^{t_r}$. Equation (24) is to balance the stocks of all the vertiports, where $A_{ji}^{t_r-1}$ is the actual number of eVTOLs arrived at vertiport i during the period of $t_r - 1$, and $D_{ij}^{t_r-1}$ means the actual number of eVTOLs departed from vertiport i during the period of $t_r - 1$. Equation (25) defines the solution domain.

$$\min_{RL_{ij}^{t_r}} c_2 \cdot \sum_{ij} ec_{r,ij} \cdot RL_{ij}^{t_r}, \forall t_r; \quad (23)$$

$$\sum_j A_{ji}^{t_r-1} - \sum_j D_{ij}^{t_r-1} = \sum_j RL_{ij}^{t_r} - \sum_j RL_{ji}^{t_r}, \forall i, t_r; \text{ and} \quad (24)$$

$$RL_{ij}^{t_r} \in \mathbb{N}^{\geq 0}, \forall i, j, t_r. \quad (25)$$

We conduct 1,000 Monte Carlo simulations, each of which is a daily UAM operation. To accurately mimic real-world demand patterns, we generated the arrival times of each passenger using an underdispersed Conway-Maxwell Poisson distribution, characterized by a mean $\lambda_{ij}^t R_i$ and a mean-to-variance, β for each simulation. Fig. 12 illustrates a typical scenario, showing the cumulative number of passengers traveling from vertiport 1 to vertiport 2, segmented by their arrival times at vertiport 1.

The pattern depicted underscores a pronounced influx during morning and evening peak periods, reflecting higher passenger arrivals at these times compared to off-peak hours. Additionally, our simulations incorporate stochastic electric consumption, modeled by a normal distribution with specified means $ec_{s,ij}^t$ and $ec_{r,ij}$, and a mean-to-standard deviation, α .

Fig. 13 shows an instance of an arbitrarily chosen simulation. At 17:30, the states and locations of the operating eVTOLs are graphically depicted. For example, at Vertiport 2, two eVTOLs are waiting to be charged since all the chargers are occupied currently, and six eVTOLs, completely charged, are idle and waiting for the next service or relocation trips.

The system failure rate is defined as the ratio of demands with waiting times exceeding the maximum allowable time to the total demand. We observe an overall service failure rate of 0.93%, with an average failure rate of 1.24% during peak hours (17:00–19:00). Fig. 14 presents a boxplot of the service failure rates for all six operating time intervals. The mean values are marked as “x”, and the boxes indicate the 25–75 percentile values. The whiskers are the ranges of the 2.5–97.5 percentile values. During three time windows—11:00–13:00, 13:00–15:00, and 15:00–17:00—most of the failure rates from 1000 simulations are zero. However, the failure rates for a few simulations are positive, which makes the mean values above zero.

Note that, in the optimal planning procedure, the average failure rate in the worst time period was set to $1 - q_1 = 2.5\%$. The agent-based simulation results show that the average failure rates for all the time periods are lower than the set level of 2.5%. Even in the worst case, i.e., time period of 9:00–11:00, the average failure rate is lower than 2.5% since we have newly introduced a real-time demand-responsive relocation strategy in the simulations. Due to this further flexible operation, the level of service increases in terms of the lower service failure rate, but the average rate of total electric costs increases by 4.40% compared to the estimated value at the planning level.

In conclusion, the agent-based Monte Carlo simulations confirm that the proposed optimal planning, derived from an analytical model rather than computationally intensive numerical optimization methods, is robust against uncertainties of demand and electric consumption based on the resulting failure rates, which can be one of the most important serviceability measures.

6. Practical implementation

This section discusses practical implementation challenges and insights related to the optimization framework for UAM systems. The discussion focuses on interaction with other transportation modes, regulatory compliance, and integration with existing urban infrastructure, all of which are critical for ensuring the feasibility and scalability of

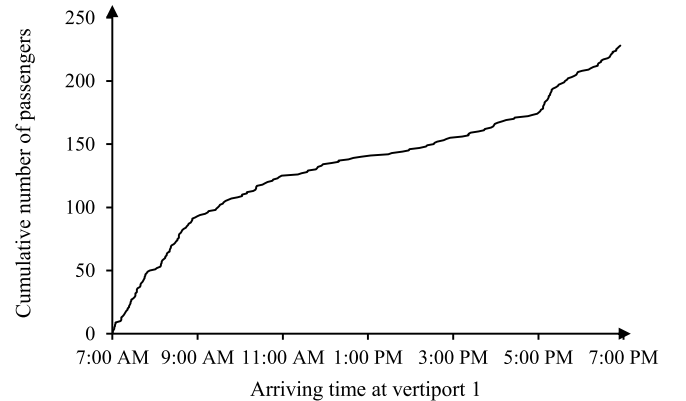


Fig. 12. Cumulative number of passengers distribution according to arrival time.

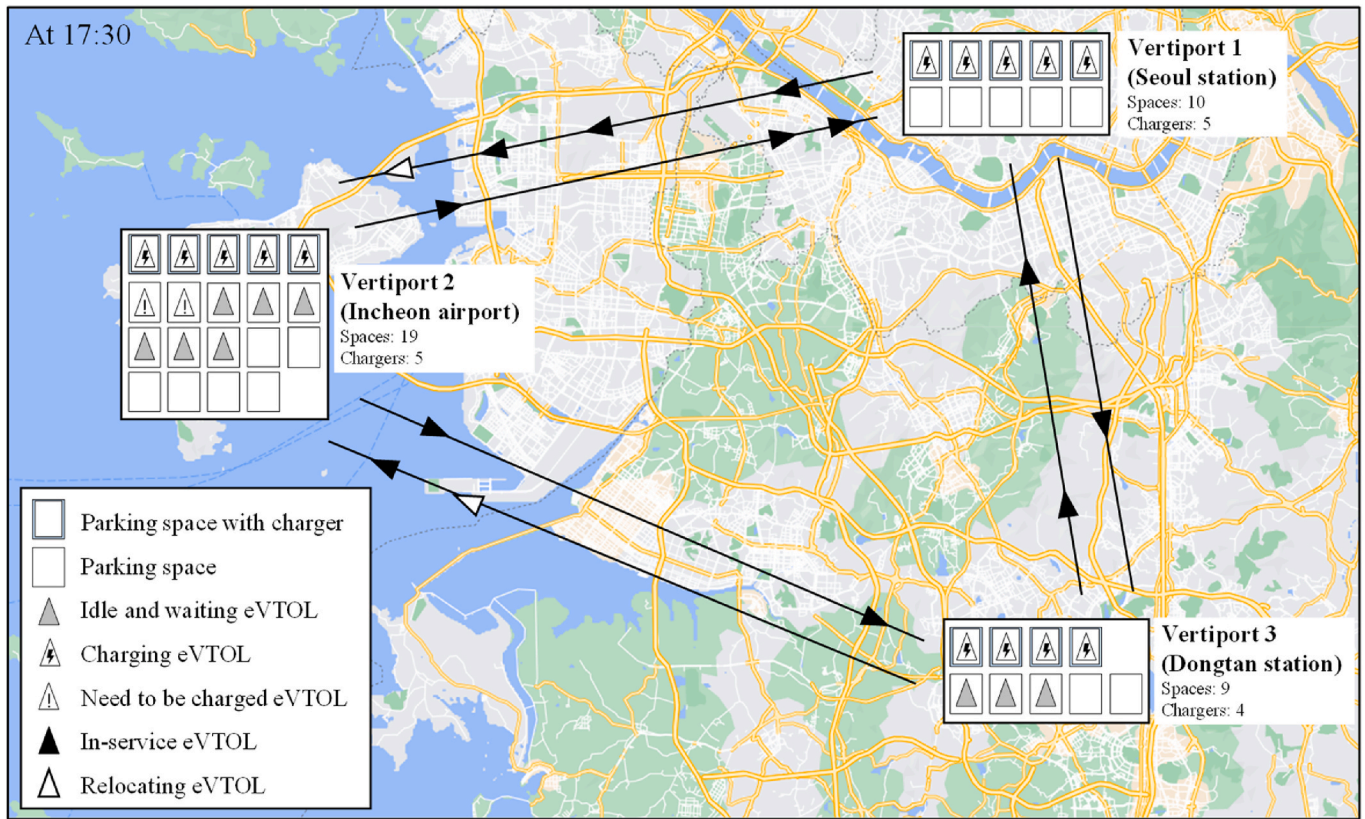


Fig. 13. System status example (at 17:30 for an arbitrarily chosen simulation).

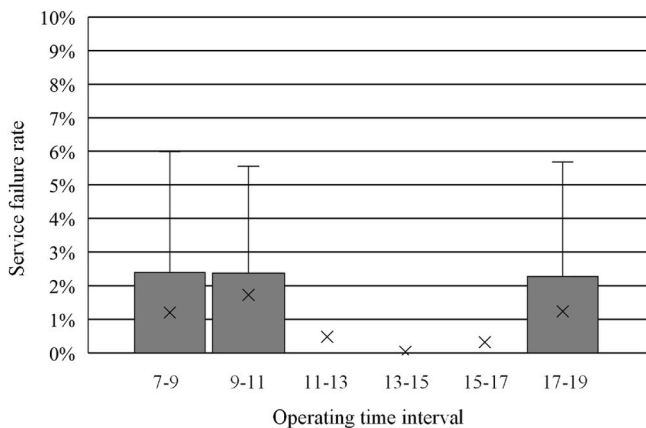


Fig. 14. Boxplot of service failure rates for each operating time interval.

UAM operations in real-world urban environments.

6.1. Integration with other transportation modes

The integration of UAM with existing transportation systems is essential for resolving first- and last-mile connectivity challenges. Since eVTOLs cannot directly reach passengers' doorsteps, their efficiency depends on how seamlessly they connect with ground transportation options, such as public transit, ride-hailing services, and micromobility solutions (e.g., bikes, scooters). Failure to provide seamless integration could reduce the potential time savings that UAM promises. Moreover, the adoption of UAM can be significantly enhanced by providing multimodal ticketing systems and real-time data integration, allowing passengers to plan their journeys efficiently across multiple transport

modes (Shon et al., 2024; Jiang et al., 2024).

Our framework can be expanded to integrate shuttle systems for first- and last-mile connections. In this expanded model, not only do we account for the planning of eVTOL operations, but we also optimize shuttle routes, schedules, and allocations to maximize system efficiency. A multimodal optimization framework could be developed, wherein both eVTOL and shuttle services are planned together to minimize total travel time, reduce passenger transfer time, and enhance overall user experience. This holistic approach will ensure that the proposed UAM system operates efficiently across all journey segments, ultimately improving user satisfaction.

6.2. Regulatory compliance

Regulatory compliance is another significant challenge for UAM deployment, as different regions have varying regulations governing airspace usage, vertiport construction, and environmental standards. The regulatory framework is evolving as UAM becomes a reality. In regions like the United States, the Federal Aviation Administration (FAA) has already launched the UAM Grand Challenge to facilitate the safe and efficient integration of eVTOLs into the national airspace system (FAA, 2023). Nevertheless, aligning operational capabilities with these regulations will require ongoing collaboration between UAM operators, urban planners, and policymakers.

One crucial aspect of regulatory compliance involves establishing safety standards for eVTOL operations, including takeoff and landing procedures near densely populated areas. Noise regulations and environmental impact assessments must also be addressed to gain public acceptance for UAM operations in urban centers. Additionally, the challenges associated with securing airspace for commercial eVTOL operations will need to be resolved, particularly in areas with high air traffic volumes. These regulations should also be incorporated into our optimization framework as constraints, ensuring that our proposed

solutions align with the evolving regulatory landscape. By integrating these constraints, our framework can provide more realistic and compliant operational strategies for UAM deployment.

6.3. Integration with existing urban infrastructure

A successful UAM system requires the strategic integration of vertiports with existing urban infrastructure. Vertiports should be located near transportation hubs such as metro stations, airports, and bus stations to facilitate easy access for passengers. The framework assumes that vertiports will act as key intermodal hubs, enabling passengers to transfer between different transportation modes efficiently. This approach not only ensures convenient connectivity but also optimizes urban space utilization by leveraging existing infrastructure. However, challenges remain regarding the location of vertiports in congested urban areas. Space availability, public opposition to vertiport construction in residential areas, and the cost of real estate will affect where vertiports can be built. Vertiport construction near key transportation hubs will require cooperation with city planners and transportation agencies to ensure that they align with broader urban development plans.

Our presented framework accounts for the availability of existing urban infrastructure and the associated costs of space usage in the optimization process. However, in this study, we limit the number and location of vertiports to simplify the problem. Future expansions of this framework could involve optimizing both the number and locations of vertiports dynamically to better adapt to urban development needs.

7. Conclusion

In this paper, we presented an optimal decision-support model for a UAM system. The planning decision variables included the number of eVTOLs, chargers, and parking spaces at each vertiport and the operational variables to flight frequency for relocation trips and the number of idle eVTOLs. The objective function aimed to minimize the total operation cost while maintaining the LOS and resilience against demand and electricity consumption stochasticity. Our solution, considering the unique properties of UAM systems, decomposed the MIP-based formulation into resilient steady-state subproblems for multiple discrete time windows.

A numerical study was conducted using three vertiport locations in the Seoul Metropolitan Area, South Korea. The results of the study showed that vertiport operation costs account for nearly half of the total agency expenses. Additionally, we conducted extensive sensitivity analyses, which revealed key insights into the relationships between operating costs and several factors, such as maximum waiting time for passengers at vertiports, charging speed, and cruising speed. Through Monte Carlo simulations, we confirmed the robustness of our planning and operational strategies against stochastic variations in demand and electric consumption.

In future research, we will explore how to refine and practically implement the model by conducting case studies that estimate real and reliable data inputs, ensuring the methodology's adaptability to diverse urban settings. Additionally, future study will take into account various operational features that can be introduced into UAM services, such as ridesharing, Mobility-as-a-Service (MaaS)-based reservations, and transfers. Particularly, ground transportation modes for first- and last-mile trips of UAM trip chain could be jointly optimized with UAM systems. Furthermore, even if the present study considered only economic factors, other important societal values, including equity and climate change mitigation, can be pursued at the same time in the process of planning and operating UAM systems.

CRedit authorship contribution statement

Heeseung Shon: Conceptualization, Data curation, Formal analysis,

Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Jinwoo Lee:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The author declares no conflict of interest regarding the publication of this paper.

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Data availability

Data will be made available on request.

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